

Research Manager

Security Fundamentals Laboratory,
Cybersecurity Research Institute,
National Institute of Information and Communications Technology.

4-2-1, Nukui-Kitamachi, Koganei, Tokyo 184-8795, Japan

EDUCATION

The University of Osaka

Ph.D. in Engineering, Advisor: Prof. Atsuko Miyaji

Suita, Osaka

Apr. 2017–May. 2019

- Thesis: A Study on Statistical Cryptanalysis of Stream Ciphers

Japan Advanced Institute of Science and Technology

Master's degree in Information Science, Advisor: Prof. Atsuko Miyaji

Nomi, Ishikawa

Apr. 2013–Mar. 2015

National Defense Academy

Bachelor's degree in Engineering, Advisor: Prof. Shigeru Yanagi, Prof. Nobuyuki Tamura

Yokosuka, Kanagawa

Apr. 2005–Mar. 2009

WORK EXPERIENCE

National Institute of Information and Communications Technology

Research Manager/Security Fundamentals Laboratory, Cybersecurity Research Institute
Senior Researcher/Security Fundamentals Laboratory, Cybersecurity Research Institute

Koganei, Tokyo

Apr. 2026–present

Apr. 2020–Mar. 2026

- Studies on Cryptanalysis of Symmetric-Key Cryptography
- Studies on Design of Symmetric Key Ciphers
- Studies on Security Analysis of Real-world Systems

The University of Osaka

Guest Associated Professor/Graduate School of Information Science and Technology

Suita, Osaka

Apr. 2025–present

- Studies on Design and Cryptanalysis of Symmetric-Key Cryptography

Toho University

Part-time Lecturer/Faculty of Science, Department of Information Science

Narashino, Chiba

Apr. 2025–present

- Social Informatics

Japan Air Self-Defense Force (Kokujeitai), Ministry of Defense

Information and Communications Officer

Shinjuku, Tokyo

Apr. 2009–Mar. 2020

AWARDS

- | | |
|---|-----------|
| ▪ ICSS Research Award, ICSS group of IEICE | Jun. 2026 |
| ▪ CSS Best Paper Award, CSS 2025, IPSJ | Oct. 2025 |
| ▪ SCIS Innovation Paper Award, SCIS 2024, ISEC group of IEICE | Jan. 2025 |
| ▪ CSS Paper Award, CSS 2024, IPSJ | Oct. 2024 |

▪ CSS Ethical Practice Award , CSS 2024, IPSJ	Oct. 2024
▪ CSS Student Paper Award , CSS 2023, IPSJ	Oct. 2023
▪ JIP Specially Selected Paper , IPSJ	Sep. 2023
▪ IPSJ Yamashita SIG Research Award , IPSJ	Mar. 2023
▪ SCIS Innovation Paper Award , SCIS 2021, ISEC group of IEICE	Jan. 2022
▪ CSS Encouragement Research Award , CSS 2021, CSEC group of IPSJ	Oct. 2021
▪ SCIS Paper Award , SCIS 2015, ISEC group of IEICE	Jan. 2016
▪ Best Student Award , Japan Advanced Institute of Science and Technology	Mar. 2015
▪ Best Student Award , National Defense Academy	Mar. 2009

COMMITTEES

- **Editorial Board:** ToSC 2025/2026, ToSC 2024/2025, ToSC 2023/2024, ToSC 2022/2023
- **Program Committees:** FSE 2026, FSE 2025, ISC 2024, ACISP 2024, FSE 2024, ADSC 2024, FSE 2023, ADSC 2023
- **Organizing Committees:** IWSEC 2024, ACNS 2023, ISPEC 2022, ACNS 2021

PEER-REVIEWED JOURNAL ARTICLES

1. Kodai Taiyama, **Ryoma Ito**, Kosei Sakamoto, Kazuma Taka, Keita Toyama, and Takanori Isobe. Generalizing Bit-based Rebound Attacks on AES-like Hashing. *IACR Communications in Cryptology*, Vol. 3, Issue 3, Oct. 2026.
2. Subhadeep Banik, Tatsuya Ishikawa, Takanori Isobe, **Ryoma Ito**, Kazuhiko Minematsu, Kazuma Nakata, Mostafizar Rahman, and Kosei Sakamoto. Dialga: A Family of Low-Latency Tweakable Block Ciphers using Multiple Linear Layers. *IACR Transactions on Symmetric Cryptology*, Vol. 2025, Issue 4, pp. 70-124, Dec. 2025.
3. Takoro Shiraya, Subhadeep Banik, Tatsuya Ishikawa, **Ryoma Ito**, Mostafizar Rahman, Kosei Sakamoto, Atsushi Tanaka, Shion Utsumi, and Takanori Isobe. Exploring SHA Instructions and Its Application to AES-based Schemes. *IACR Communications in Cryptology*, Vol. 2, Issue 3, Oct. 2025.
4. Kosei Sakamoto, **Ryoma Ito**, and Takanori Isobe. Parallel SAT framework to find clustering of differential characteristics and its applications. *Journal of Information Security and Applications*, vol. 94, no. 104203, Sep. 2025 (IF = 3.7).
5. Kazuma Taka, Kosei Sakamoto, **Ryoma Ito**, Rentaro Shiba, Shion Utsumi, and Takanori Isobe. Divide-and-Conquer SAT for Exploring Optimal Differential and Linear Characteristics and Its Applications. *IACR Transactions on Symmetric Cryptology*, Vol. 2025, Issue 3, pp. 516–576, Sep. 2025.
6. Hayato Watanabe, **Ryoma Ito**, and Toshihiro Ohigashi. On the Effects of Neural Network-based Output Prediction Attacks on the Design of Symmetric-key Ciphers. *Journal of Information Security and Applications*, vol. 90, no. 104046, May 2025 (IF = 3.8).
7. Takanori Isobe, **Ryoma Ito**, and Kazuhiko Minematsu. Security Analysis of SFrame. *Journal of Information Security and Applications*, vol. 89, no. 103958, Mar. 2025 (IF = 3.8).
8. Yu Long Chen, Antonio Flórez-Gutiérrez, Akiko Inoue, **Ryoma Ito**, Tetsu Iwata, Kazuhiko Minematsu, Nicky Mouha, Yusuke Naito, Ferdinand Sibleyras, and Yosuke Todo. Key Committing Security of AEZ and More. *IACR Transactions on Symmetric Cryptology*, Vol. 2023, Issue 4, pp. 452–488, Dec. 2023.
9. Takanori Isobe, **Ryoma Ito**, and Kazuhiko Minematsu. Cryptanalysis on End-to-End Encryption Schemes of Communication Tools and Its Research Trend (Invited Paper). *Journal of Information Processing*, vol. 31, pp. 523-536, Sep. 2023.
10. Hayato Kimura, Keita Emura, Takanori Isobe, **Ryoma Ito**, Kazuto Ogawa, and Toshihiro Ohigashi. A Deeper Look into Deep Learning-based Output Prediction Attacks Using Weak SPN Block Ciphers. *Journal of Information Processing*, vol. 31, pp. 550-561, Sep. 2023.

11. Nasratullah Ghafoori, Atsuko Miyaji, **Ryoma Ito**, and Shotaro Miyashita. PNB Based Differential Cryptanalysis of Salsa20 and ChaCha. *IEICE Trans. Information and Systems*, vol. E106-D, no.9, pp. 1407-1422, Sep. 2023.
12. Takanori Isobe, **Ryoma Ito**, Fukang Liu, Kazuhiko Minematsu, Motoki Nakahashi, Kosei Sakamoto, and Rentaro Shiba. Areion: Highly-Efficient Permutations and Its Applications of Hash Functions for Short Input. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, Vol. 2023, Issue 2, pp. 115-154, Mar. 2023.
13. Akinori Hosoyamada, Akiko Inoue, **Ryoma Ito**, Tetsu Iwata, Kazuhiko Minematsu, Ferdinand Sibleyras, and Yosuke Todo. Cryptanalysis of Rocca and Feasibility of Its Security Claim. *IACR Transactions on Symmetric Cryptology*, Vol. 2022, Issue 3, pp. 123-151, Sep. 2022.
14. Fukang Liu, Willi Meier, Santanu Sarkar, Gaoli Wang, **Ryoma Ito**, and Takanori Isobe. New Cryptanalysis of ZUC-256 Initialization Using Modular Differences. *IACR Transactions on Symmetric Cryptology*, Vol. 2022, Issue 3, pp. 152-190, Sep. 2022.
15. Jin Hoki, Takanori Isobe, **Ryoma Ito**, Fukang Liu, and Kosei Sakamoto. Distinguishing and Key Recovery Attacks on the Reduced-Round SNOW-V and SNOW-Vi. *Journal of Information Security and Applications*, vol. 65, no. 103100, Mar. 2022. (IF = 4.960).
16. Takanori Isobe and **Ryoma Ito**. Security Analysis of End-to-End Encryption for Zoom Meetings. *IEEE Access*, vol. 9, pp. 90677-90689, Jun. 2021 (IF = 3.367).
17. **Ryoma Ito**, Rentaro Shiba, Kosei Sakamoto, Fukang Liu, and Takanori Isobe. Bit-wise Cryptanalysis on AND-RX Permutation Friet-PC. *Journal of Information Security and Applications*, vol. 59, no. 102860, Jun. 2021 (IF = 3.872).
18. **Ryoma Ito** and Atsuko Miyaji. New Iterated RC4 Key Correlations and Their Application to Plaintext Recovery on WPA-TKIP. *IEICE Trans. Fundamentals*, vol. E104-A, no. 1, pp. 190-202, Jan. 2021.
19. **Ryoma Ito** and Atsuko Miyaji. Refined Construction of RC4 Key Setting in WPA. *IEICE Trans. Fundamentals*, vol. E100-A, no. 1, pp. 138-148, Jan. 2017.
20. **Ryoma Ito** and Atsuko Miyaji. Refined RC4 Key Correlations of Internal States in WPA. *IEICE Trans. Fundamentals*, vol. E99-A, no. 6, pp. 1132-1144, Jun. 2016.
21. **Ryoma Ito** and Atsuko Miyaji. Refined Glimpse Correlations of RC4. *IEICE Trans. Fundamentals*, vol. E99-A, no. 1, pp.3-13, Jan. 2016.

PEER-REVIEWED INTERNATIONAL CONFERENCE PAPERS

1. Tatsuya Ishikawa, Takanori Isobe, **Ryoma Ito**, Kosei Sakamoto, Kosuke Sasaki, Kazuma Taka, and Shion Utsumi. Finding Collision for HMAC-SHA Forgery in the Secret-State Setting. *Advanced in Cryptology – Asiacrypt 2026*, volume XXXXX of Lecture Notes in Computer Science, pages XXX-XXX, 2026.
2. Kento Morita, Tatsuya Ishikawa, Takuro Shiraya, Kosei Sakamoto, **Ryoma Ito**, and Takanori Isobe. Accelerated Differential Search for Gleeok Using Hamming-Weight-Guided SAT Solving. *International Conference on Cryptology and Network Security – CANS 2026*, volume XXXXX of Lecture Notes in Computer Science, pages XXX-XXX, 2026.
3. Kota Tatsumi, Hayato Watanabe, **Ryoma Ito**, and Toshihiro Ohigashi. Refining the Evaluation of Diffusion Property for Simon Variants. *International Symposium on Information Theory and Its Applications – ISITA 2026*, pages XXX-XXX, IEEE, 2026.
4. Keita Toyama, Kosei Sakamoto, **Ryoma Ito**, Kazuma Taka, Kodai Taiyama, and Takanori Isobe. Automated Tool for Finding Practical Collisions on SPN-Based Hash Functions. *Australasian Conference on Information Security and Privacy – ACISP 2026*, volume 16792 of Lecture Notes in Computer Science, pages 3-39, 2026.
5. Taichi Nagoya, Takuro Shiraya, Kazuma Taka, Tatsuya Ishikawa, Kosei Sakamoto, **Ryoma Ito**, and Takanori Isobe. Analyzing Forgery Security of LeMac: Tight Bounds and Impact of Padding. *Australasian Conference on Information Security and Privacy – ACISP 2026*, volume 16792 of Lecture Notes in Computer Science, pages 79-101, 2026.

6. Takehiro Matsumoto, Atsushi Tanaka, Kyosuke Yamashita, **Ryoma Ito**, and Takanori Isobe. Comprehensive Security Analysis of End-to-End Encryption in LINE. *International Conference on Information Security and Cryptography – ICISC 2025*, volume 16487 of Lecture Notes in Computer Science, pages 237–256, 2025.
7. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, and Takanori Isobe. Gravity of the Situation: Security Analysis on Rocket.Chat E2EE. *IEEE Annual Computer Security Applications Conference – ACSAC 2025*, pages 15–30, 2025.
8. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, Shogo Shiraki, and Takanori Isobe. Not in The Prophecies: Practical Attacks on Nostr. *IEEE European Symposium on Security and Privacy – EuroS&P 2025*, pages 585–606, 2025.
9. Kodai Taiyama, Kosei Sakamoto, **Ryoma Ito**, Kazuma Taka, and Takanori Isobe. Key Collisions on AES and Its Applications. *Advanced in Cryptology – Asiacrypt 2024*, volume 15490 of Lecture Notes in Computer Science, pages 267–300, 2024.
10. Hayato Watanabe, **Ryoma Ito**, and Toshihiro Ohigashi. On the Effects of Neural Network-based Output Prediction Attacks on the Design of Symmetric-key Ciphers. *Cyber Security, Cryptology, and Machine Learning – CSCML 2024*, volume 15349 of Lecture Notes in Computer Science, pages 201–218, 2024.
11. Kosei Sakamoto, **Ryoma Ito**, and Takanori Isobe. Parallel SAT Framework to Find Clustering of Differential Characteristics and Its Applications. *Selected Area in Cryptography – SAC 2023*, volume 14201 of Lecture Notes in Computer Science, pages 409–428, 2023.
12. Hayato Kimura, Keita Emura, Takanori Isobe, **Ryoma Ito**, Kazuto Ogawa, and Toshihiro Ohigashi. Output Prediction Attacks on Block Ciphers using Deep Learning. *Artificial Intelligence and Industrial Internet-of-Things Security – AIoT 2022*, volume 13285 of Lecture Notes in Computer Science, pages 248–276, 2022.
13. Shotaro Miyashita, **Ryoma Ito**, and Atsuko Miyaji. PNB-focused Differential Cryptanalysis of ChaCha Stream Cipher. *Australasian Conference on Information Security and Privacy – ACISP 2022*, volume 13494 of Lecture Notes in Computer Science, pages 46–66, 2022.
14. Keita Emura, **Ryoma Ito**, Sachiko Kanamori, Ryo Nojima, and Yohei Watanabe. State-free End-to-End Encrypted Storage and Chat Systems based on Searchable Encryption. In *Proceedings of the 24th International Conference on Enterprise Information Systems – Volume 2: ICEIS*, SCITEPRESS, pages 106–113, 2022.
15. Takanori Isobe, **Ryoma Ito**, and Kazuhiko Minematsu. Security Analysis of SFrame. *European Symposium on Research in Computer Security – ESORICS 2021*, volume 12973 of Lecture Notes in Computer Science, pages 127–146, 2021.
16. Takanori Isobe and **Ryoma Ito**. Security Analysis of End-to-End Encryption for Zoom Meetings. *Australasian Conference on Information Security and Privacy – ACISP 2021*, volume 13083 of Lecture Notes in Computer Science, pages 234–253, 2021.
17. Jin Hoki, Takanori Isobe, **Ryoma Ito**, Fukang Liu, and Kosei Sakamoto. Distinguishing and Key Recovery Attacks on the Reduced-Round SNOW-V. *Australasian Conference on Information Security and Privacy – ACISP 2021*, volume 13083 of Lecture Notes in Computer Science, pages 171–190, 2021.
18. **Ryoma Ito**. Rotational Cryptanalysis of Salsa Core Function. *International Conference on Information Security – ISC 2020*, volume 12472 of Lecture Notes in Computer Science, pages 129–145, 2020.
19. **Ryoma Ito** and Atsuko Miyaji. New Iterated RC4 Key Correlations. *Australasian Conference on Information Security and Privacy – ACISP 2018*, volume 10946 of Lecture Notes in Computer Science, pages 154–171, 2018.
20. **Ryoma Ito** and Atsuko Miyaji. How TKIP Induces Biases of Internal States of Generic RC4. *Australasian Conference on Information Security and Privacy – ACISP 2015*, volume 9144 of Lecture Notes in Computer Science, pages 329–342, 2015.
21. **Ryoma Ito** and Atsuko Miyaji. New Linear Correlations Related to State Information of RC4 PRGA Using IV in WPA. *Fast Software Encryption – FSE 2015*, volume 9054 of Lecture Notes in Computer Science, pages 557–576, 2015.
22. **Ryoma Ito** and Atsuko Miyaji. New Integrated Long-Term Glimpse of RC4. *Information Security Application – WISA 2014*, volume 8909 of Lecture Notes in Computer Science, pages 137–149. 2015.

OTHER INTERNATIONAL CONFERENCE PRESENTATIONS

1. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, and Takanori Isobe. Payload Compromised: Full Key Recovery in Rocket.Chat E2EE. *Black Hat Asia 2026 Briefings*, Apr. 2026.
2. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, Shogo Shiraki, and Takanori Isobe. Not Sealed: Practical Attacks on Nostr, a Decentralized Censorship-Resistant Protocol. *Black Hat USA 2025 Briefings*, Aug. 2025.
3. Yu Long Chen, Antonio Flórez-Gutiérrez, Akiko Inoue, **Ryoma Ito**, Tetsu Iwata, Kazuhiko Minematsu, Nicky Mouha, Yusuke Naito, Ferdinand Sibleyras, and Yosuke Todo. Key Committing Security of AEZ. *The Third NIST Workshop on Block Cipher Modes of Operation 2023*, Oct. 2023.

NON PEER-REVIEWED PAPERS

1. Subhadeep Banik, Tatsuya Ishikawa, Takanori Isobe, **Ryoma Ito**, Kazuhiko Minematsu, Kazuma Nakata, Mostafizar Rahman, and Kosei Sakamoto. Dialga: A Family of Low-Latency Tweakable Block Ciphers using Multiple Linear Layers (Full Version). *IACR Cryptology ePrint Archive*: Report 2026/547, December 2025.
2. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, and Takanori Isobe. Gravity of the Situation: Security Analysis on Rocket.Chat E2EE. *IACR Cryptology ePrint Archive*: Report 2025/2300, December 2025.
3. Hayato Kimura, **Ryoma Ito**, Kazuhiko Minematsu, Shogo Shiraki, and Takanori Isobe. Not in The Prophecies: Practical Attacks on Nostr. *IACR Cryptology ePrint Archive*: Report 2025/1459, August 2025.
4. Kodai Taiyama, Kosei Sakamoto, **Ryoma Ito**, Kazuma Taka, and Takanori Isobe. Key Collisions on AES and Its Applications. *IACR Cryptology ePrint Archive*: Report 2024/1508, September 2024.
5. Hayato Watanabe, **Ryoma Ito**, and Toshihiro Ohigashi. On the Effects of Neural Network-based Output Prediction Attacks on the Design of Symmetric-key Ciphers. *IACR Cryptology ePrint Archive*: Report 2024/1310, August 2024.
6. Kosei Sakamoto, **Ryoma Ito**, and Takanori Isobe. Parallel SAT Framework to Find Clustering of Differential Characteristics and Its Applications. *IACR Cryptology ePrint Archive*: Report 2023/1227, August 2023.
7. Takanori Isobe, **Ryoma Ito**, Fukang Liu, Kazuhiko Minematsu, Motoki Nakahashi, Kosei Sakamoto, and Rentaro Shiba. Areion: Highly-Efficient Permutations and Its Applications (Extended Version). *IACR Cryptology ePrint Archive*: Report 2023/794, June 2023.
8. Shotaro Miyashita, **Ryoma Ito**, and Atsuko Miyaji. PNB-focused Differential Cryptanalysis of ChaCha Stream Cipher. *IACR Cryptology ePrint Archive*: Report 2021/1537, November 2021.
9. Fukang Liu, Willi Meier, Santanu Sarkar, Gaoli Wang, **Ryoma Ito**, and Takanori Isobe. New Cryptanalysis of ZUC-256 Initialization Using Modular Differences. *IACR Cryptology ePrint Archive*: Report 2021/1104, August 2021.
10. Keita Emura, **Ryoma Ito**, Sachiko Kanamori, Ryo Nojima, and Yohei Watanabe. State-free End-to-End Encrypted Storage and Chat Systems based on Searchable Encryption. *IACR Cryptology ePrint Archive*: Report 2021/953, July 2021.
11. Jin Hoki, Takanori Isobe, **Ryoma Ito**, Fukang Liu, and Kosei Sakamoto. Distinguishing and Key Recovery Attacks on the Reduced-Round SNOW-V and SNOW-Vi. *IACR Cryptology ePrint Archive*: Report 2021/546, April 2021.
12. Takanori Isobe and **Ryoma Ito**. Security Analysis of End-to-End Encryption for Zoom Meetings. *IACR Cryptology ePrint Archive*: Report 2021/486, April 2021.
13. Takanori Isobe, **Ryoma Ito**, and Kazuhiko Minematsu. Security Analysis of SFrame. *IACR Cryptology ePrint Archive*: Report 2021/424, April 2021.
14. Hayato Kimura, Keita Emura, Takanori Isobe, **Ryoma Ito**, Kazuto Ogawa, and Toshihiro Ohigashi. Output Prediction Attacks on Block Ciphers using Deep Learning. *IACR Cryptology ePrint Archive*: Report 2021/401, March 2021.
15. **Ryoma Ito**, Rentaro Shiba, Kosei Sakamoto, Fukang Liu, and Takanori Isobe. Bit-wise Cryptanalysis on AND-RX Permutation Friet-PC. *IACR Cryptology ePrint Archive*: Report 2021/212, March 2021.